

COURSE PROJECT
KAGGLE PREDICTION COMPETITION
EXPLORING MENTAL HEALTH DATA

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507 – DATA ANALYTICS IN ENGINEERING SYSTEMS

AGENDA

- 1. Competition Overview**
- 2. Training datasets Description - Input variables**
- 3. Iteration 1 - MATLAB hybrid model**
- 4. Iteration 8 - MATLAB stacked ensemble model**
- 5. Iteration 12 - Python CatBoost model**
- 6. Iteration 18 - Python master code with 3 Sub-model**
- 7. Final Iteration 40 - Python master code with 5 Sub-model**
- 8. Challenges and Learnings**

COMPETITION OVERVIEW



Problem Statement

Predict depression risk using demographic and lifestyle factors on competition testing dataset.

Binary Classification Problem with target variable *Depression*



Competition Dataset Features

18 input variables

1 target variable

Synthetically driven from mental health survey data [1,2]

“balance between having real-world data (with named features) and ensuring test labels are not publicly available”



Original Depression Survey Dataset

2556 observation

18 variables

1 target variable

Each observation is a volunteer with 18 features recorded and the output is the patient experiencing depression or not



Goal

Build a prediction model to explore factors that may cause or contribute more to experiencing depression

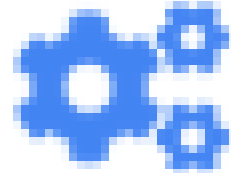
My model is training using both competition training dataset and original depression survey dataset

TRAINING DATASETS DESCRIPTION - INPUT VARIABLES

Variable	Type	Reasoning
Name	Categorical (nominal)	Unique identifier
Gender	Categorical (nominal)	Represents categories without inherent ordering
Age	Continuous	Represents a measurable, numeric value that can take on a wide range of values
City	Categorical (nominal)	Represents distinct categories for cities without order
Working Professional or Student	Categorical (nominal)	Represents two categories
Profession	Categorical (nominal)	Represents various professions without inherent order
Academic Pressure	Categorical (ordinal)	Ordinal scale from 1 to 5 indicating pressure level
Work Pressure	Categorical (ordinal)	Ordinal scale from 1 to 5 indicating pressure level
CGPA	Continuous	Represents a measurable, numeric value that can take on a wide range of values
Study Satisfaction	Categorical (ordinal)	Ordinal scale 1 to 5
Job Satisfaction	Categorical (ordinal)	Ordinal scale 1 to 5
Sleep Duration	Categorical (ordinal)	Represents categories with a natural order
Dietary Habits	Categorical (ordinal)	Represents categories with a natural order
Degree	Categorical (nominal)	Represents various types of degrees without inherent order
Have you ever had suicidal thoughts?	Categorical (nominal)	Binary categorical variable with two categories
Work/Study Hours	Continuous	Represents a measurable, numeric value that can take on a wide range of values
Financial Stress	Categorical (ordinal)	Ordinal scale 1 to 5
Family History of Mental Illness	Categorical (nominal)	Binary categorical variable with two categories

ITERATION 1

MATLAB HYBRID MODEL



Methodology

Gaussian Naïve Bayes (GNB) combined with Naïve Bayes Classifier



Challenges

Limited model accuracy (82%) due to simplistic assumptions

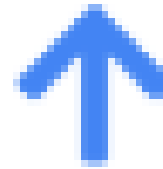
ITERATION 8

MATLAB STACKED ENSEMBLE MODEL



Advanced Methodology

Combined Random
Forest, Gradient
Boosting, kNN, and
XGBoost



Improved Accuracy

Achieved 94.2%, a
significant leap from
earlier iterations

ITERATION 12

PYTHON CATBOOST MODEL

Transition to Python for advanced modeling capabilities

Engineered additional enhanced features

1. **Pressure** → Includes work or academic pressure based on the observation status
2. **Satisfaction** → Includes job or study satisfaction
3. **PS_category**
4. **PS_ratio** → Computes the pressure to satisfaction ratio
5. **PSF_ratio** → Computes the pressure-satisfaction-financial stress interaction ratio
6. **PF_factor** → Computes the pressure-financial stress interaction
7. **Age_bin** → Bins all ages into discrete intervals

Hyperparameter optimization for CatBoost by using **Optuna**

Results: 94.3% accuracy with **5-Fold Cross-Validation**

ITERATION 18

PYTHON MASTER CODE WITH 3 SUB-MODEL



Ensemble Approach

Combined CatBoost 1 from iteration 12, CatBoost 2 (simplified preprocessing/no feature engineering), and XGBoost



Innovative Techniques

Added Majority Voting and Weighted Averaging



Outcome

Achieved 94.4% accuracy

FINAL ITERATION 40

PYTHON MASTER CODE WITH 5 SUB-MODEL



Enhanced Ensemble

Combined CatBoost 1
from iteration 12,
CatBoost 2 from iteration
18, CatBoost 2 variant 1,
CatBoost 2 variant 2, and
XGBoost



Outcome

Achieved 94.45%
accuracy



CHALLENGES AND LEARNINGS

- **Data Cleaning** → Handled inconsistencies between both training datasets and handled missing values by averaging
- **Feature Engineering** → Created innovative features that boosted the model performance
- **Model Iteration** → Balanced simplicity of preprocessing and powerfulness of modeling to avoid overfitting and “feel data”



**THANK
YOU**